ALL INDIA TEST SERIES
SELECTION HOGA YAHIN SE

Dropper JEE Main

AJPM7/06

Test- 06

DURATION : 180 Minutes

DATE : 19/01/2025

M.MARKS : 300

# ANSWER KEY

**PHYSICS**

1. (1)
2. (3)
3. (2)
4. (3)
5. (1)
6. (4)
7. (2)
8. (4)
9. (3)
10. (4)
11. (1)
12. (3)
13. (1)
14. (1)
15. (4)
16. (2)
17. (1)
18. (1)
19. (2)
20. (3)
21. (3)
22. (4)
23. (24)
24. (40)
25. (60)

**CHEMISTRY**

26. (4)
27. (3)
28. (4)
29. (4)
30. (1)
31. (3)
32. (2)
33. (2)
34. (2)
35. (2)
36. (3)
37. (1)
38. (2)
39. (2)
40. (2)
41. (4)
42. (3)
43. (2)
44. (4)
45. (3)
46. (65)
47. (3)
48. (154)
49. (150)
50. (5)

**MATHEMATICS**

51. (2)
52. (2)
53. (3)
54. (2)
55. (4)
56. (2)
57. (2)
58. (3)
59. (2)
60. (3)
61. (4)
62. (4)
63. (4)
64. (2)
65. (3)
66. (3)
67. (1)
68. (3)
69. (1)
70. (1)
71. (169)
72. (2)
73. (1)
74. (5)
75. (3)

## SECTION-I (PHYSICS)

**1.** **(1)**

It is given that, $F = -\alpha x^3$

$\Rightarrow \quad \frac{mvdv}{dx} = -\alpha x^3$

$\Rightarrow \quad \int_{v_0}^{0} v dv = -\frac{\alpha}{m}\int_0^x x^3 dx$

$\Rightarrow \quad \frac{0^2}{2} - \frac{v_0^2}{2} = \frac{-\alpha}{4m} x^4$

$\Rightarrow \quad \frac{-v_0^2}{2} = -\frac{\alpha}{4m} x^4$

$\Rightarrow \quad x = \left(\frac{2v_0^2 m}{\alpha}\right)^{1/4}$

**[24 Feb, 2021 (Shift-II)]**

**2.** **(3)**

At equilibrium, $\frac{dU}{dr} = 0 \Rightarrow \frac{d}{dr}\left[\frac{A}{r^8}\right] - \frac{d}{dr}\left[\frac{B}{r^4}\right] = 0$

$\Rightarrow \quad -8Ar^{-8-1} - B(-4)r^{-4-1} = 0$

$\Rightarrow \quad -8Ar^{-9} + 4Br^{-5} = 0$

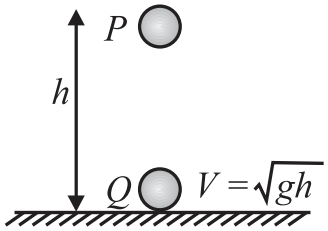
$8Ar^{-9} = 4Br^{-5}$

$\Rightarrow \quad r^4 = \frac{2A}{B}$

$\therefore \quad r = \left(\frac{2A}{B}\right)^{\frac{1}{4}}$

**[24 June, 2022 (Shift-II)]**

**3.** **(2)**




Time for collision $t = \sqrt{\frac{h}{g}}$

At the time of collision $\vec{P}_i = \vec{P}_f$

Velocity inter change due to same mass

**[8 Jan, 2020 (Shift-II)]**

**4.** **(3)**

$\rho(r) = \rho_0\left(1 - \frac{r}{R}\right)$

$\frac{dm}{dV} = \rho_0\left(1 - \frac{r}{R}\right)$

$dm = \rho_0\left(1 - \frac{r}{R}\right)dV$

$\therefore \quad m = \int_0^r \rho_0\left(1 - \frac{r}{R}\right) \times 4\pi r^2 dr$

$= 4\pi\rho_0\left(\frac{r^3}{3} - \frac{r^4}{4R^2}\right)$

$\therefore \quad E = \frac{Gm}{r^2} = \frac{G4\pi\rho_0}{r^2}\left[\frac{r^3}{3} - \frac{r^4}{4R^2}\right] = G4\pi\rho_0\left[\frac{r}{3} - \frac{r^2}{4R^2}\right]$

$\therefore \quad \frac{dE}{dr} = 0 \Rightarrow r = \frac{2R}{3}$

**[Sep, 2020]**

**5.** **(1)**

$K_1 (100 - 50) = K_2 (20)$

$K_3 (30) = K_2(20)$

**[6 Sep, 2020 (Shift-II)]**

**6.** **(4)**

$y(t) = \frac{y_0}{2}(1 - \cos 4\omega t)$

Amplitude $\frac{Mg}{k} = \frac{y_0}{2}$

$\Rightarrow \quad \frac{k}{M} = \frac{2g}{y_0}$

Angular frequency $4\omega = \sqrt{\frac{k}{m}}$

$\Rightarrow \quad \omega = \frac{1}{4}\sqrt{\frac{k}{m}} = \frac{1}{4}\sqrt{\frac{2g}{y_0}} = \sqrt{\frac{g}{8y_0}}$

**[2 Sep, 2020 (Shift-II)]**

**7.** **(2)**

Here,

(*K.E.* + *P.E.*)initial = (*K.E.* + *P.E.*)final

$0 + \left(\frac{1}{4\pi \in_0}\frac{2P}{8a^3}\right) \times P = 2 \times \frac{1}{2}mv^2 + 0$

$v = \sqrt{\frac{P^2}{16\pi \in ma^3}} = \frac{P}{4a}\sqrt{\frac{1}{\pi \in ma}}$

**[6 Sep, 2020 (Shift-II)]**

**8.** **(4)**

$T \sin\theta = 2qE$ .....(i)

$T \cos\theta = mg$ .....(ii)

On dividing (i) by (ii) we get,

$\tan\theta = \frac{2qE}{mg}$

$$Q = \frac{C_1C_2}{C_1 + C_2}(V_1 + V_2)$$

$$\theta = \tan^{-1}\left[\frac{2qE}{mg}\right]$$

$$\Rightarrow \quad \theta = \tan^{-1}\left[\frac{2q}{mg} \times \frac{KC_2(V_1 + V_2)}{(C_1 + C_2)t}\right]$$

**[27 July, 2021 (Shift-II)]**

**9. (3)**

Maximum current through galvanometer, $I_{max}$
$= 50 \times 20 \times 10^{-6} = 10^{-3}$ A

(i) $I_{max} = \frac{4}{100 + R_1} = 10^{-3} \Rightarrow R_1 = 3900\Omega$

(ii) $I_{max} = \frac{20}{R_G + R_1 + R_2}$

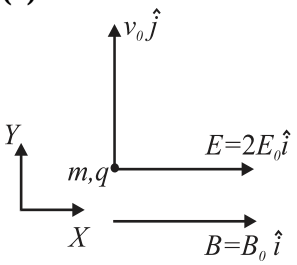
$$10^{-3} = \frac{20}{100 + 3900 + R_2} \Rightarrow R_2 = 16000\Omega$$

(iii) $I_{max} = \frac{40}{100 + 3900 + 16000 + R_3}$

$R_3 = 20000\Omega$

**[12 April, 2019(Shift-I)]**

**10. (4)**




$(2v_0)^2 = v_0^2 + v_x^2; v_x = \sqrt{3}v_0$

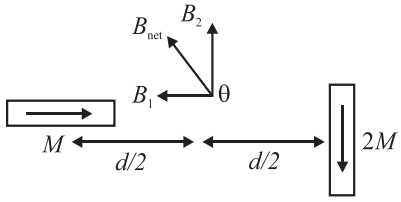
$\therefore \quad \sqrt{3}v_0 = 0 + \frac{2qE_0}{m}t \Rightarrow; t = \frac{mv_0\sqrt{3}}{2qE_0}$

**[17 Jan, 2020 (Shift-II)]**

**11. (1)**

For dipole *X*:

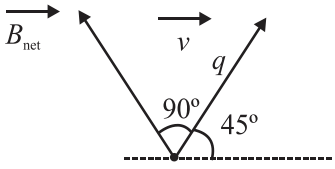
$$B_x = 2\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3}$$




For dipole *Y*:

$\because \quad B_1 = B_2$

$\therefore \quad B_{net}$ is at 45° from the horizontal line




Velocity of charge q at 45° from the horizontal line.

Force on charge, $F = q(\vec{v} \times \vec{B}) =$ maximum

**[8 April, 2019 (Shift-II)]**

**12. (3)**

$$q = \int I dt$$

$$q = \int_0^{2L/R} \frac{E}{R}\left[1 - e^{\frac{-Rt}{L}}\right]dt$$

$$\Rightarrow \quad q = \frac{EL}{R^2}\left[1 + \frac{1}{e^2}\right]$$

**[12 April, 2019 (Shift-II)]**

**13. (1)**

The capacitor gets repeatedly charged and discharged due to alternating square wave source

**During Charging Process:**

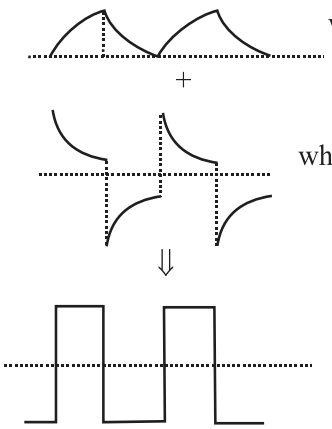
$Q = CV_0\left(1 - e^{\frac{-t}{RC}}\right) \Rightarrow$ Charge on the capacitor, and

$V_C = \frac{Q}{C} = V_0\left(1 - e^{\frac{-t}{RC}}\right) \Rightarrow$ Potential difference across capacitor

**During discharging Process.**

$$Q = Q_0 e^{\frac{1}{RC}} + CV_0\left(1 - e^{\frac{1}{RC}}\right)$$

$\Rightarrow$ Charge on the capacitor




**[16 March, 2021 (Shift-I)]**

**14. (1)**

Direction of propagation $\hat{s} = \frac{2\hat{k} - 3\hat{j}}{\sqrt{13}}$

**[21 April, 2019 (Shift-I)]**

**15. (4)**

At focus, magnification is ∞

**[8 Jan, 2020 (Shift-II)]**

**16. (2)**

$$\sqrt{(2d)^2+(d)^2}-2d=\lambda$$

$$\Rightarrow \quad (\sqrt{5}-2)d=\lambda$$

$$\Rightarrow \quad d=\frac{\lambda}{(\sqrt{5}-2)}$$

**[19 Jan, 2019(Shift-II)]**

**17. (1)**

$$\lambda_0=\frac{h}{\frac{m}{2}v_0}$$

$$\frac{m}{2}v_0=\frac{m}{2}v_1+mv_2$$

$$v_0=v_1+2v_2$$

$$v_1=\frac{-v_0}{3}$$

$$\lambda_1=\frac{h}{\frac{m}{2}\frac{v_0}{3}}=3\lambda_0$$

$$\Delta\lambda=2\lambda_0$$

**18. (1)**

$$\frac{t_2}{t_1}=\frac{P_1}{P_2}\times\sqrt{\frac{T_2}{T_1}}$$

$$\frac{t_2}{t_1}=\frac{P}{2P}\times\sqrt{\frac{900}{300}}$$

$$t_2=3\times10^{-8}\times\sqrt{3}\simeq3\sqrt{3}\times10^{-8}\ \text{sec}$$

**[12 Jan, 2019 (Shift-II)]**

**19. (2)**

Radius $R = RA^{1/3} \Rightarrow R \propto A^{1/3}$

$V \propto R^3$

$\therefore\ V \propto A$

And mass $\propto A$

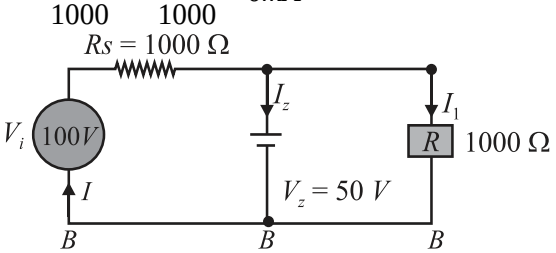
So, density is independent of $A$

**[29 July, 2022 (Shift-II)]**

**20. (3)**

$$I_2=\frac{V_i-V_z}{R_S}-\frac{V_z}{R}$$

$$=\frac{100-50}{1000}-\frac{50}{1000}=0mA$$




**[20 July, 2021 (Shift-I)]**

**21. (3)**

Both pipes vibrate in first overtone

$$f_0=\frac{1}{\sqrt{\rho_1}}\frac{1}{L}$$

$$f_c=\frac{1}{\sqrt{\rho_2}}\frac{1}{\frac{4L'}{3}}$$

$$f_0=f_c$$

$$L'=\frac{3L}{4}\sqrt{\frac{\rho_1}{\rho_2}}$$

**[16 March, 2021 (Shift-II)]**

**22. (4)**

Given, $\omega = x\sqrt{F}$ rad/s

We know that, $F=m\omega^2\frac{L}{2}$

$$\Rightarrow \quad F=\frac{mL}{2}\left(x^2F\right)\Rightarrow x^2=\frac{2}{mL}=\frac{2}{\frac{1}{4}\times\frac{1}{2}}=16$$

$\therefore\ x = 4$

**[29 July, 2022 (Shift-II)]**

**23. (24)**

$a = -\mu_k g = -2$

$V = 12 - 2 \times 2$

$V = 8$ m/s

$$KE=\frac{1}{2}mv^2+\frac{1}{2}\frac{mr^2}{2}\frac{v^2}{r^2}=\frac{3}{4}mv^2$$

$KE = 3 \times 8 = 24$ J

**[30 Jan, 2023 (Shift-II)]**

**24. (40)**

$$\varepsilon=\left[\left|\frac{d\phi}{dt}\right|\right]=A\frac{dB}{dt}=\pi\times1^2\times\frac{4}{\pi}\times10^{-3}\times\frac{1}{100}=4\times10^{-5}V$$

When magnetic field switched off, $B = 0$,

$$\therefore \quad \frac{4}{\pi}\times10^{-3}\left(1-\frac{t}{100}\right)=0$$

$\Rightarrow \quad t = 100$ second

Energy dissipated,

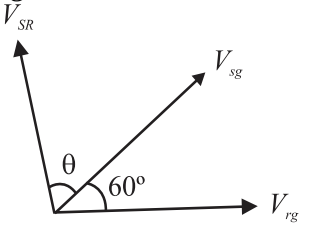
$$E=I^2Rt=\frac{\varepsilon^2}{R}t=\frac{(4\times10^{-5})^2}{4\times10^{-6}}\times100=40\ mJ.$$

**[25 July, 2021 (Shift-I)]**

**25. (60)**

Velocity vector is of same magnitude. Therefore, resultant would pass exactly midway through them So, $\theta = 60^o$

Hence, angle $\theta$ with the line $AB$ is $60^o$




**[27 July, 2021 (Shift-II)]**

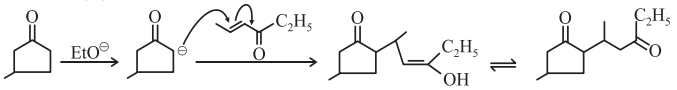
## SECTION-II (CHEMISTRY)

**26. (4)**

Paper chromatography is a type of partition chromatography.

**[28 July, 2022 (Shift-II)]**

**27. (3)**



**[9 Jan, 2020 (Shift-I)]**

**28. (4)**

All statements are correct

**[6 April, 2023 (Shift-II)]**

**29. (4)**

$$\text{Mole fraction of } H_2SO_4 = \frac{\left(\frac{49}{98}\right)}{\left(\frac{49}{98}\right)+\left(\frac{51}{18}\right)}$$

$= 0.15.$

**[9 Jan, 2020 (Shift-I)]**

**30. (1)**

$E^\circ_{cell}$ (standard electrode potential)

$(E°)_{option\ 1} = (2.16)\ V$

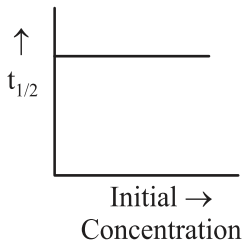
$(E°)_{option\ 2} = (1.24)\ V$

$(E°)_{option\ 3} = (1.56)\ V$

$(E°)_{option\ 4} = (1.84)\ V$

**[11 Jan, 2019 (Shift-I)]**

**31. (3)**




$t_{1/2}$ of 1st order independent of initial concentration.

**[25 July, 2021 (Shift-I)]**

**32. (2)**

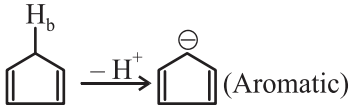
$$O_2\left(16e^-\right) = \sigma_{1s^2}\sigma^*_{1s^2}\ \sigma_{2s^2}\sigma^*_{2s^2}\ \sigma_{2p_z{}^2}\pi_{2p_x{}^2} =$$

$$\pi_{2p_y{}^2}\pi^*_{2p_x{}^1} = \pi^*_{2p_y{}^1}\sigma^*_{2p_z}$$

LUMO = $\sigma^*_{2p_z}$

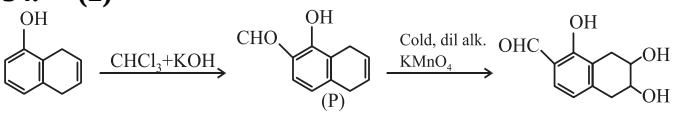
**[6 Sept, 2020 (Shift-I)]**

**33. (2)**




**[31 Aug, 2021 (Shift-I)]**

**34. (2)**




**35. (2)**

$Cr^{3+} = [Ar]3d^3$

$\mu_{spin-only} = \sqrt{n(n+2)}$ B.M

$\mu = 3.87$

$\mu = 38.7 \times 10^{-1}$

$Z = 39$

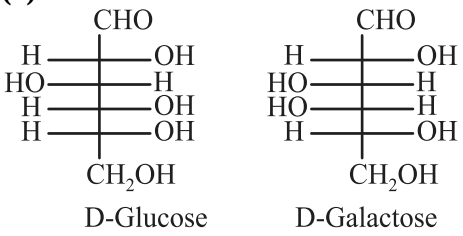
**[17 March, 2021 (Shift-II)]**

**36. (3)**

1-(S), 2-(R), 3-(P), 4-(Q)

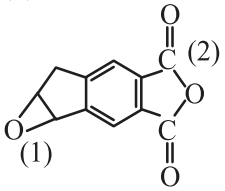
**[25 Jan, 2023 (Shift-II)]**

**37. (1)**




**[6 Sept, 2020 (Shift-II)]**

**38. (2)**




**[11 April, 2023 (Shift-I)]**

**39. (2)**

$$n = \frac{2.7099\times10^{24}}{6.022\times10^{23}}$$

$n = 4.5$

Weight of $NH_3 = 4.5 \times 17 = 76.5$ g

Volume at STP = 100.8 L

No. of atoms = $1.084 \times 10^{25}$ atoms

**[10 April, 2023 (Shift-I)]**

**40. (2)**

$$\lambda_0 = \frac{c}{\nu_0}$$

$$\lambda_0 = \frac{3.0\times10^8\ m/s}{2.0\times10^{15}\ s^{-1}}$$

$\lambda = 1.5 \times 10^{-7}$ m

**[25 June, 2022 (Shift-II)]**

**41. (4)**

Metallic character increases top to bottom & hence basic character of oxides increases and acidic character decreases

**[31 Aug, 2021 (Shift-I)]**

**42. (3)**




$d^2sp^3$, inner orbital complex, low spin and octahedral

**[2 Sept, 2020 (Shift-I)]**

**43. (2)**

In an adiabatic system, no heat is exchanged with the surroundings. The energy from the combustion reaction remains within the system, leading to an increase in the internal energy of the system.

Since the system is isolated and no heat can escape, the energy from the combustion process is converted into the thermal energy of the system. Therefore, **the temperature in System X will increase.**

In a diathermic system, the container allows heat exchange with the surroundings.

When propane undergoes combustion, the system can transfer heat to the surroundings. If the system is in thermal equilibrium with its surroundings, the temperature remains constant.

**[13 April, 2023 (Shift-II)]**

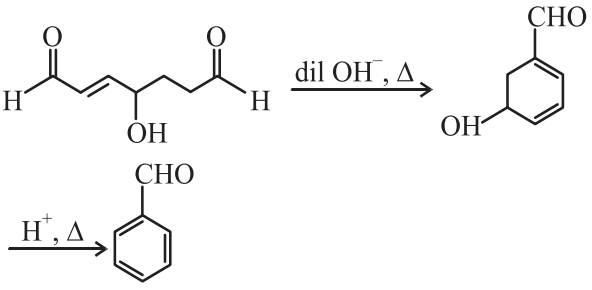
**44. (4)**

Oxidation state of sulfur in $SO_4^{2-}$ is +6 and the oxidation state of sulfur in $SO_2$ is +4.

Therefore the difference in oxidation state of sulfur in sulfate ion ($SO_4^{2-}$) and sulfur dioxide ($SO_2$) is +2.

**[24 June, 2022 (Shift-I)]**

**45. (3)**




**[27 Aug, 2021 (Shift-I)]**

**46. (65)**

$$K_{sp} = \left[P^{2+}\right]^2\left[Q^{2-}\right]$$

Since $\left[P^{2+}\right] = 2S_P$ and $\left[Q^{2-}\right] = S_P$, we have:

$$K_{sp} = \left(2S_P\right)^2 \cdot S_P = 4S_P^3$$

Given $K_{sp} = 5.0\times10^{-9}$, we solve for $S_P$:

$$4S_P^3 = 5.0\times10^{-9}$$

$$S_P^3 = \frac{5.0\times10^{-9}}{4} = 1.25\times10^{-9}$$

$$S_P = \sqrt[3]{1.25\times10^{-9}} \approx 1.08\times10^{-3}\text{M}$$

$$K_{sp} = \left[R^{+}\right]^3\left[S^{3-}\right]$$

Since $\left[R^{+}\right] = 3S_R$ and $\left[S^{3-}\right] = S_R$, we have :

$$K_{sp} = \left(3S_R\right)^3 \cdot S_R = 27S_R^4$$

Given $K_{sp} = 2.0\times10^{-6}$, we solve for $S_R$:

$$27S_R^4 = 2.0\times10^{-6} \quad S_R^4 = \frac{2.0\times10^{-6}}{27} \approx 7.41\times10^{-8}$$

$$S_R = \sqrt[4]{7.41\times10^{-8}} \approx 1.65\times10^{-2}\text{M}$$

$$\frac{S_P}{S_R} = \frac{1.08\times10^{-3}}{1.65\times10^{-2}} \approx 0.065$$

x = 0.065

1000x = 65

**[16 March, 2021 (Shift-I)]**

**47. (3)**

Complex formula: $[FeCl(H_2O)_4(NH_3)]Cl_2$.

Oxidation state of Fe = +3

No. of Cl atoms coordinated to Fe = 1.

$$\frac{3x+6y}{5} = 3$$

**[27 July, 2022 (Shift-I)]**

**48. (154)**

$Cu^{+2} + 2e^- \rightarrow Cu \quad E^\circ = +0.34v \quad \Delta G_1$

$Cu^{+} + e^- \rightarrow Cu \quad E^\circ = +0.52V \quad \Delta G_2$

$Cu^{2+} + e^- \rightarrow Cu^{+} \quad \Delta G_3$

$\Delta G_3 = \Delta G_1 - \Delta G_2$

$=(-2 \times 96{,}000 \times 0.34) - (-1 \times 96{,}000 \times 0.52)$

$= -65{,}280 + 49{,}920$

$\Delta G_3 = -15{,}360\text{J} = -154 \times 10^2$

**[10 April, 2023 (Shift-I)]**

**49.** **(150)**

$$\ln\left(\frac{k_2}{k_1}\right)=\frac{E_a}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

$$\ln\left(\frac{0.9}{1.8\times10^{-4}}\right)=\ln(5000)\approx 8.517$$

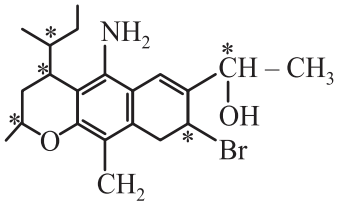
$$8.51=\frac{E_a}{8.31}\times\left(\frac{1}{593}-\frac{1}{823}\right)$$

$$E_a=\frac{8.51\times8.31\times593\times823}{230}\text{ J/mol}$$

$$E_a\approx 150057\text{ J/mol}$$

**[10 April, 2019 (Shift-I)]**

**50.** **(5)**

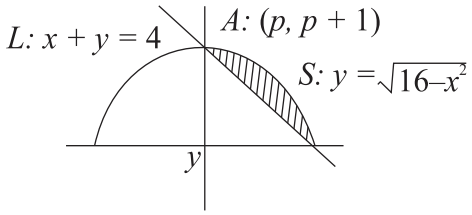



**[2 Sept, 2020 (Shift-I)]**

## SECTION-III (MATHEMATICS)

**51.** **(2)**

$$4-x\le y\le\sqrt{16-x^2};0\le x\le 4$$




$$L(A)>0\Rightarrow p+p+1-4\ge 0\Rightarrow p\ge\frac{3}{2}$$

……(i)

$$S(A)<0\Rightarrow p+1-\sqrt{16-p^2}\le 0$$

$$\Rightarrow p+1\le\sqrt{16-p^2}$$

$$\Rightarrow p^2+2p+1\le 16-p^2$$

$$\Rightarrow 2p^2+2p-15\le 0$$

$$\Rightarrow p\in\left[\frac{-1-\sqrt{31}}{2},\frac{-1+\sqrt{31}}{2}\right]\qquad\text{……(ii)}$$

From eqn (i) $\cap$ eqn(ii), we get

$$p\in\left[\frac{3}{2},\frac{\sqrt{31}-1}{2}\right]\equiv[a,b]$$

**[06 April, 2023 (Shifts-I)]**

**52.** **(2)**

$\because$ $\alpha,\beta$ be the roots of $x^2+70^{\frac{1}{4}}x+a=0$

$\therefore\ \alpha+\beta=-70^{\frac{1}{4}}$ & $\alpha\beta=a$

Given $\alpha^4+\beta^4=-30\Rightarrow\left(\alpha^2+\beta\right)^2-2\alpha^2\beta^2=-30$

$$\Rightarrow\left\{(\alpha+\beta)^2-2\alpha\beta\right\}^2-2a^2=-30$$

$$\Rightarrow\left\{70^{\frac{1}{2}}-2a\right\}^2-2a^2=-30$$

$$\Rightarrow 70+4a^2-4a\times70^{\frac{1}{2}}-2a^2=-30$$

$$\Rightarrow 2a^2-4.70^{\frac{1}{2}}a+100=0$$

$$\Rightarrow a^2-2\sqrt{70}a+50=0$$

Product of roots = 50

**[25 Jan, 2023 (Shifts-II)]**

**53.** **(3)**

$$\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+\frac{1}{(60-a)(80-a)}$$

$$+.....+\frac{1}{(380-a)(400-a)}=\frac{1}{256}$$

LHS

$$=\frac{1}{20}\left(\frac{1}{20-a}-\frac{1}{40-a}\right)+\frac{1}{20}\left(\frac{1}{40-a}-\frac{1}{60-a}\right)+$$

$$\cdots\frac{1}{20}\left(\frac{1}{380-a}-\frac{1}{400-a}\right)$$

$$=\frac{1}{20}\left(\frac{1}{20-a}-\frac{1}{400-a}\right)=\frac{1}{20}.\frac{380}{(20-a)(400-a)}$$

$$\frac{19}{(20-a)(400-a)}=\frac{1}{256}\text{ (given)}$$

$\therefore$ Product of roots = 3136.

**[29 July, 2022 (Shifts-I)]**

**54.** **(2)**

Given set $A=\{a,b,c\}$

Relation $R$ on set $A$

$R=\{(a,a)(a,b)(a,c)(b,a)(b,b)(b,c)(c,b)(c,a)(c,c)\}$

For symmetric $(a,b)\in R\Rightarrow(b,a)\in R$

Hence $R$ is symmetric

For transitive $(a,b)\in R,(b,c)\in R\Rightarrow(a,c)\in R$

Hence R is transitive

So element must be added on $R=\{(a,a)(a,c)(b,a)(b,b)(c,a)(c,b)(c,c)\}$

So number of element = 7

**[30 Jan, 2023 (Shifts-I)]**

**55. (4)**

Given that $\bar{x} = 50 \Rightarrow \sum x_i = 500$

$\sum x_{i_{correct}} = 500 + 15 + 25 - 40 - 50 = 45$

$\sigma^2 = 144$

$\frac{\sum x_i^2}{10} - (50)^2 = 144$

$\sum x_{i_{correct}}^2 = (144 + (50)^2) \times 10 - (40)^2 - (50)^2 + (15)^2$

$+ (25)^2 = 23190$

Correct variance = $\frac{\sum (x_{icorrect})^2}{10} - \left(\frac{\sum x_{icorrect}}{10}\right)^2$

$= 2319 - (45)^2$

$= 2319 - 2025 = 294$

**[13 April, 2023 (Shifts-I)]**

**56. (2)**

Given $\left|\frac{z-4i}{z+i}\right| = 4$

$\Rightarrow (z-4i)(\bar{z}+4i = 16(z+i)(\bar{z}-i)$

$\Rightarrow z\bar{z} + 16 + 4i(z-\bar{z}) = 16(z\bar{z} + 1 + i(\bar{z}-z))$

$\Rightarrow 15z\bar{z} - 20i(z-\bar{z}) = 0$

Put $z = x + iy$ and $\bar{z} = x - iy$

$\Rightarrow 3(x^2 + y^2) - 4i(2iy) = 0$

$\Rightarrow x^2 + y^2 + \frac{8}{3}y = 0$

This is a equation of circle where centre $(0, -\frac{4}{3})$

**[25 Jan, 2023 (Shifts-II)]**

**57. (2)**

NALANDA

$\Rightarrow$ AAADLNN

$A \rightarrow \frac{6!}{2!2!} = 180$

$D \rightarrow \frac{6!}{3!2!} = 60$

$L \rightarrow \frac{6!}{3!2!} = 60$

$NA\underline{A} \rightarrow 4! = 24$

$NA\underline{D} \rightarrow \frac{4!}{2!} = 12$

$NALA\underline{A} \rightarrow 2!$

$NALA\underline{D} \rightarrow 2!$

$NALAN\,\underline{A} \rightarrow 1$

$NALANDA \rightarrow 1$

Rank of NALANDA = 342

**[6 April, 2023 (Shifts-II)]**

**58. (3)**

Let $P(2\lambda + 1, \lambda + 3, 2\lambda + 2)$

Let $Q = (\mu + 2, 2\mu + 2, 3\mu + 3)$

Directions ratios of $PQ$ will be parallel to $(1, -1, 2)$

$\Rightarrow \frac{2\lambda - \mu - 1}{1} = \frac{\lambda - 2\mu + 1}{-2} = \frac{2\lambda - 3\mu - 1}{-2}$

$\Rightarrow \lambda = -7, \mu = -9$

$P(-13, -4, -12)$

$Q(-7, -16, -24) \Rightarrow PQ = 18.$

**[25 Jan, 2023 (Shifts-I)]**

**59. (2)**

General term = $\frac{10!}{r_1! r_2! r_3!}(a)^{r_1}(bx)^{r_2}(cx^2)^{r_3}$

$(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$

$P_1$ = Coefficient of $x = 20$

For coefficient of $x : r_2 + 2r_3 = 1$

$\Rightarrow$ $\begin{matrix} r_1 & r_2 & r_3 \\ 9 & 1 & 0 \end{matrix}$

$20 = \frac{10!}{9!1!} \times a^9 \times b^1$

$a^9.b = 2$

$a = 1, b = 2$

Coefficient of $x^2 = 360$

$\Rightarrow$ $\begin{matrix} r_1 & r_2 & r_3 \\ 9 & 0 & 1 \end{matrix}$ or $\begin{matrix} r_1 & r_2 & r_3 \\ 8 & 2 & 0 \end{matrix}$

$360 = \frac{10!}{9!1!} \times a^9 \times c^1 + \frac{10!}{8!2!} \times a^8 b^2$

$360 = 10c + 45 \times 4$

$\Rightarrow 10c = 180$

$\Rightarrow c = 18$

$\therefore (a + b + c) = 21$

**[15 April, 2023 (Shifts-I)]**

**60. (3)**

Length of Latus Rectum = $\frac{2b^2}{a} = \frac{1}{2}$

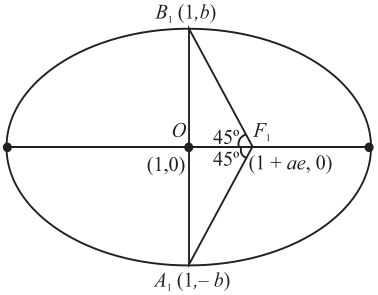
$\Rightarrow 4b^2 = a$ .....(i)

Ellipse $\frac{(x-1)^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{b}{ae} = \tan 45^\circ$

$\Rightarrow \frac{b}{ae} = 1 \Rightarrow b^2 = a^2e^2$

$\Rightarrow a^2e^2 = a^2 - b^2$



$\Rightarrow 2b^2 = a^2$

From (i) and (i)

$a = 2a^2$

$\therefore\ a = \dfrac{1}{2}$

$\Rightarrow b^2 = \dfrac{1}{8} \Rightarrow 4b^2 = \dfrac{1}{2}$

Hence $((2a)^2 + (2b)^2) = \dfrac{3}{2}$

**[15 April, 2023 (Shifts-I)]**

**61. (4)**

Let $P(H) = x$ then $P(T) = 1 - x$

$\Rightarrow\ P(4H1T) = P(3H2T)$

$\Rightarrow\ {}^5C_4(x)^4(1-x)^1 = {}^5C_3x^3(1-x)^2$

$\Rightarrow\ 10(1-x) = 5x$

$\Rightarrow\quad x = \dfrac{10}{15} = \dfrac{2}{3}$

$P$(atmost 2$H$)

$= P(0H,5T) + P(1H,4T) + P(2H,3T)$

$= {}^5C_0\left(\dfrac{1}{3}\right)^5 + {}^5C_1\dfrac{2}{3}.\left(\dfrac{1}{3}\right)^4 + {}^5C_2\left(\dfrac{2}{3}\right)^2\left(\dfrac{1}{3}\right)^3$

$= \dfrac{1}{3^5}(1+10+40) = \dfrac{51}{243}$

**[26 Jan, 2022 (Shifts-I)]**

**62. (4)**

$|A - \lambda I| = 0$

$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ -2 & -5-\lambda \end{vmatrix} = 0 \Rightarrow (\lambda-1)(\lambda+5)+4 = 0$

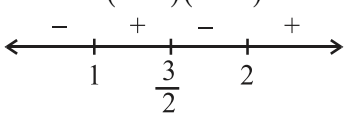
$\Rightarrow \lambda_2 + 4\lambda - 1 = 0$

$\Rightarrow A^2 + 4A = I$

**[27 July, 202 (Shifts-I)]**

**63. (4)**

$f'(x) = \dfrac{(2x-3)}{(x-1)(x-2)} > 0$




$x \in \left(1, \dfrac{3}{2}\right) \cup (2, \infty)$ but $x \in \left(1, \dfrac{3}{2}\right)$ not in domain

**[28 July, 2022 (Shifts-I)]**

**64. (2)**

$3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0$

$3\cos^2 2\theta + 6\cos 2\theta - \dfrac{10(1+\cos 2\theta)}{2} + 5 = 0$

$3\cos^2 2\theta + \cos 2\theta = 0$

$\cos 2\theta = 0$ or $\cos 2\theta = -1/3$

$\theta \in \left(0, \dfrac{\pi}{2}\right)$ only 2 solution

**[29 Jan, 2022 (Shifts-I)]**

**65. (3)**

Use $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$,

$\cos(\sin^{-1}x) = \sqrt{1-x^2}$

$\tan^{-1}\sqrt{1-x^2} = \cot^{-1}\dfrac{1}{\sqrt{1-x^2}}$

$\cot\left(\tan^{-1}\sqrt{1-x^2}\right) = \dfrac{1}{\sqrt{1-x^2}}$

$\sin^{-1}\dfrac{x}{\sqrt{1-x^2}} = \cos^{-1}\dfrac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$;

$\cos\left(\sin^{-1}\dfrac{x}{\sqrt{1-x^2}}\right) = \dfrac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$

$\Rightarrow\ \dfrac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k \Rightarrow\ x^2 = \dfrac{k^2-1}{k^2-2}\quad x = \pm\sqrt{\dfrac{k^2-1}{k^2-2}}$ ;

i.e.

$\dfrac{1}{\alpha^2} + \dfrac{1}{\beta^2} = 2\left(\dfrac{k^2-2}{k^2-1}\right); \dfrac{\alpha}{\beta} = -1$

Sum of roots $\Rightarrow b = -1 + 2\dfrac{(k^2-2)}{k^2-1}$

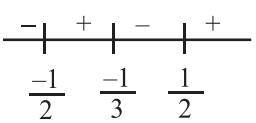
Product of roots $\Rightarrow -5 = -2\left(\dfrac{k^2-2}{k^2-1}\right)$

$b = 4, k^2 = \dfrac{1}{3}$

**[27 Jan, 2022 (Shifts-I)]**

**66. (3)**

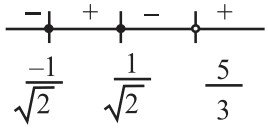
$\dfrac{6x^2+5x+1}{2x-1} > 0 \Rightarrow \dfrac{(3x+1)(2x+1)}{2x-1} > 0$




$x \in \left(\dfrac{-1}{2}, -\dfrac{1}{3}\right) \cup \left(\dfrac{1}{2}, \infty\right)$ ....(i)

and $-1 \le \dfrac{2x^2-3x+4}{3x-5} \le 1$

$\Rightarrow \dfrac{2x^2-1}{3x-5} \ge 0\ \&\ \dfrac{2x^2-6x+9}{3x-5} \le 0$




And $3x - 5 < 0$

$x \in \left[\dfrac{-1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}\right] \cup \left(\dfrac{5}{3}, \infty\right)$ .....(ii)

$x < \dfrac{5}{3} \Rightarrow x \in \left(-\infty, \dfrac{5}{3}\right)$ .....(iii)

Taking intersection of equ. (i), (ii) and (iii), we get $\left(\frac{-1}{2},-\frac{1}{3}\right)\cup\left(\frac{1}{2},\frac{1}{\sqrt{2}}\right]$. Also it is equal to $(\alpha,\beta)\cup(\gamma,\delta]$

So, $27\left(\frac{\alpha^2+\beta^2}{\gamma^2+\delta}\right)=27\left(\frac{\frac{1}{4}+\frac{1}{9}}{\frac{1}{4}+\frac{1}{2}}\right)$

$=27\cdot\frac{13}{4\cdot 9}\cdot\frac{4}{3}=13$

**[8 April, 2023 (Shifts-I)]**

**67. (1)**

Given $\alpha=\lim\limits_{x\to\pi/4}\frac{\tan^3 x-\tan x}{\cos(x+\pi/4)}$

$=\lim\limits_{x\to\pi/4}\frac{3\tan^2 x.\sec^2 x-\sec^2 x}{-\sin(x+\pi/4)}$

$\alpha=-4$

Given $\beta=\lim\limits_{x\to 0}(\cos x)^{\cot x}$

$\beta=e^{\lim\limits_{x\to 0}\left(\frac{\cos x-1}{\tan x}\right)}$, $\beta=e^{\lim\limits_{x\to 0}\frac{-\sin x}{\sec^2 x}}$

$\beta=e^0=1$

Equation whose roots are $\alpha$ and $\beta$

$x^2+3x-4=0$

**[31 Aug, 2023 (Shifts-II)]**

**68. (3)**

Given function is $f(x)=(|x-1|\sin|x-1|)$ $\cos|x-2|+|(x-3)(x-1)(x-4)|$

Cleary, $(|x-1|\sin|x-1|)$ is differentiable at $x = 1$.

And $|(x-1)(x-3)(x-4)|$ is non differentiable at $x = 1, 3, 4$

So, there are 3 points of non-differentiability.

**[29 July, 2022 (Shifts-I)]**

**69. (1)**

$n(A\cap B) = 70 + 60 - 100 = 30\%$

**[5 Nov, 2020 (Shifts-I)]**

**70. (1)**

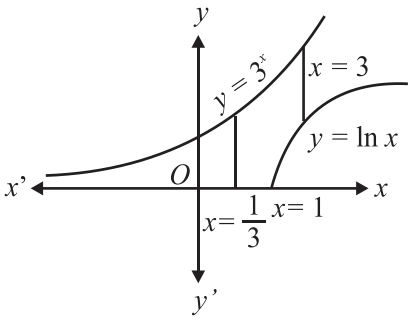
$\int\limits_0^{2\pi}\sin x dx=-\cos x\int\limits_0^{2\pi}=-(1-1)=0$

$\int\limits_0^{\pi}\sin x dx=2$

**[31 Jan, 2021 (Shifts-II)]**

**71. (169)**

$R=\left\{(x,y):\max\{0,\log_e x\}\le y\le 3^x,\frac{1}{3}\le x\le 3\right\}$




Area = $\int_{\frac{1}{3}}^{3}3^x dx-\int_1^3\ln x dx$

$=\left[\frac{3^x}{\ln 3}\right]_{1/3}^{3}-x[\ln x-1]_1^3$

$=\frac{(3^3)-3^{1/3}}{\log_2 2}-(3\ln 3-2)$

$=\frac{\alpha}{\ln 3}+\beta\log_e 3+\gamma$

$\therefore\ \alpha=3^3-\sqrt[3]{3},\beta=-3,\gamma=2$

Hence $(\alpha+2\beta-4\gamma+\sqrt[3]{3})^2$

$=\left(27-\sqrt[3]{3}-6-8+\sqrt[3]{3}\right)^2$

$=13^2=169$

**[27 July, 2021 (Shifts-I)]**

**72. (2)**

$(2y^2)y.dx+2x(x^2-y^2)dy=0$

$\Rightarrow\frac{dy}{dx}=\frac{y(-2y^2)}{2x(x^2-y^2)}$

Put $y = vx$

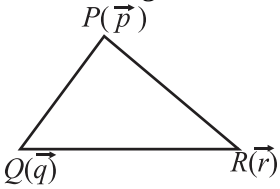
$\Rightarrow\ v+x.\frac{dv}{dx}=\frac{v(-2v^2)}{2(1-v^2)}$

$\frac{y^2}{8}-\ln y=\frac{1}{2}$

**73. (1)**

Centroid divides orthocentre and circumcentre in 2: 1.

Let $A$ be origin



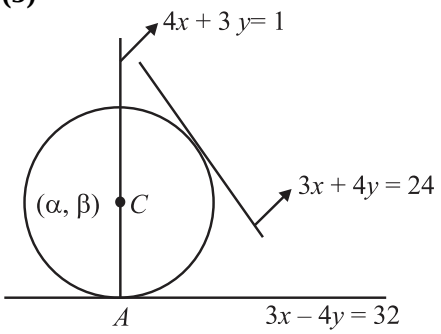

$\overrightarrow{AP}+\overrightarrow{AQ}+\overrightarrow{AR}=\vec{p}+\vec{q}+\vec{r}$

$\overrightarrow{AG}=\frac{\vec{p}+\vec{q}+\vec{r}}{3}$ (where G is centroid)

$\Rightarrow\vec{p}+\vec{q}+\vec{r}=3\overrightarrow{AG}=\overrightarrow{AB}$

**[10 April, 2023 (Shifts-II)]**

**74.** **(5)**




For point $A$ solve there equations $4x + 3y = 1$ and $3x - 4y = 32$

We , get point $A$ is $(4, -5)$

Centre $(\alpha,\beta)$ lies on $4x + 3y = 1$

$$\therefore\ 4\alpha + 3\beta = 1 \Rightarrow \beta = \frac{1-4\alpha}{3}$$

The distance from centre of circle to the lines $3x - 4y - 32 = 0$ and $3x + 4y - 24 = 0$ are equal

$$\left|\frac{3\alpha - 4\left(\frac{1-4\alpha}{3}\right) - 32}{\sqrt{3^2 + (-4)^2}}\right| = \left|\frac{3\alpha + 4\left(\frac{1-4\alpha}{3}\right) - 24}{\sqrt{3^2 + 4^2}}\right|$$

After solving $\alpha = 1$ and $\alpha = \frac{28}{3}$ (Rejected)

For $\alpha = 1$, centre $(1,-1) \Rightarrow$ radius = 5

$\Rightarrow$ radius $\approx$ 59.78 (rejected)

**[13 April, 2023 (Shifts-II)]**

**75.** **(3)**

$= (49)^{1011} + (9)^{1011}$

$= (50 - 1)^{1011} + (10 - 1)^{1011}$

$= k_1 - 1 + 5k_2 - 1$

$= 5m - 2$

Remainder = 5 – 2 = 3

**[28 July, 2022 (Shift -I)]**

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